ZOONOTIC DISEASES

Backyard zoonoses: The roles of companion animals and peri-domestic wildlife

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The spillover of human infectious diseases from animal reservoirs is now well appreciated. However, societal and climate-related changes are affecting the dynamics of such interfaces. In addition to the disruption of traditional wildlife habitats, in part because of climate change and human demographics and behavior, there is an increasing zoonotic disease risk from companion animals. This includes such factors as the awareness of animals kept as domestic pets and increasing populations of free-ranging animals in peri-domestic environments. This review presents background and commentary focusing on companion and peri-domestic animals as disease risk for humans, taking into account the human-animal interface and population dynamics between the animals themselves.

INTRODUCTION

"One Health" is an integrated, unifying approach to balance and optimize the health of people, animals, and the environment. The need for a One Health approach to limit cross-species spillover and the risk of emerging zoonoses originating from wild or domestic animals is now widely appreciated. Within this context, One Health often groups domestic animals together. Here, we follow the lead of the field of zooarcheology, which differentiates between livestock, beasts of burden, and pets. The latter two categories comprise companion animals, where the risk of zoonotic spillover may be quite distinct compared with that for livestock through the nature of the human-animal contacts. Many companion animal species have received comparatively little attention as a zoonotic risk, and societal changes in recent years are changing the dynamics of the human-animal interface. Another consideration in One Health dynamics involves peri-domestic animals and the consequences of changing land use, notably increased urbanization, which reshapes the definition of what may be considered "wildlife" (*1*).

Pathogen transmission happens when there is direct or indirect contact between an infected and a susceptible host. Whether interspecies contacts happen and whether a given species is susceptible to a given pathogen result from a complex set of ecological and molecular factors and include spillover infections from animal hosts to humans (*2*). The role of bats and rodents, as well as the role of arthropod vectors, has been reviewed extensively elsewhere. Here, we focus on companion and peri-domestic animal species as sources of zoonotic infection, including many examples where disease dynamics are currently most fluid.

Ecological context

For zoonotic pathogens, ecological dynamics within the animal reservoir and at human-animal interfaces are key drivers of spillover risk (*2*, *3*). Companion animals and peri-domestic wildlife occupy a special position in these ecological networks because of their position at the human-animal interface or the indoor-outdoor interface. These animals can play critical roles in zoonotic spillover by enabling the maintenance of a zoonotic pathogen, facilitating its spatial spread, acting as a bridge between otherwise unconnected species, or providing particular opportunities for its evolution.

Cats (*Felis catus*) present a conspicuous example of a risk in the context of zoonotic pathogen dynamics (*4*). Cats are widely kept as pets; according to the American Veterinary Medical Association, there are an estimated 60 million animals in the United States alone, often in low-density environments and in indoor-only situations. However, increasingly large populations of "unowned" feral cats are emerging around the world, often in peri-domestic environments. There are populations of cats that have both indoor-outdoor lifestyles and are "owned," as well as large numbers of cats housed with other animals in shelters or in rescue or breeding facilities, often in high-density environments; these cats are then typically adopted and become owned either permanently or temporarily. These complex dynamics need to be taken into consideration when assessing the risk of emerging zoonoses (*5*, *6*). Although less extreme, other companion animal species such as dogs (*Canis lupus*) and horses (*Equus caballus*) have equivalent nuances based on their respective position at the human-animal interface.

Similarly, bird (class Aves) species that comprise a risk for zoonotic disease encompass different categories, including pet birds, wild birds, and poultry; the latter includes birds in large-scale production facilities, as well as the increasing number of backyard poultry flocks. Classical examples of avian zoonoses include pathogens that infect people working or living in close contact with animals with overall low incidence in humans, such as transmission of *Chlamydia psittaci* from pet parrots to humans (*7*). However, there is growing interest in the role of wild birds and poultry in the dynamics of pathogens posing a wider threat to public health, notably in light of the ongoing avian influenza panzootic.

Companion animals or peri-domestic wildlife can act as notable reservoirs for pathogens that may affect human health (Fig. 1, A to C). They can also act as a bridge between wildlife and humans. The proximity of companion animals and peri-domestic wildlife with humans represents opportunities for pathogen transmission. This is particularly true for animals kept as pets or for backyard

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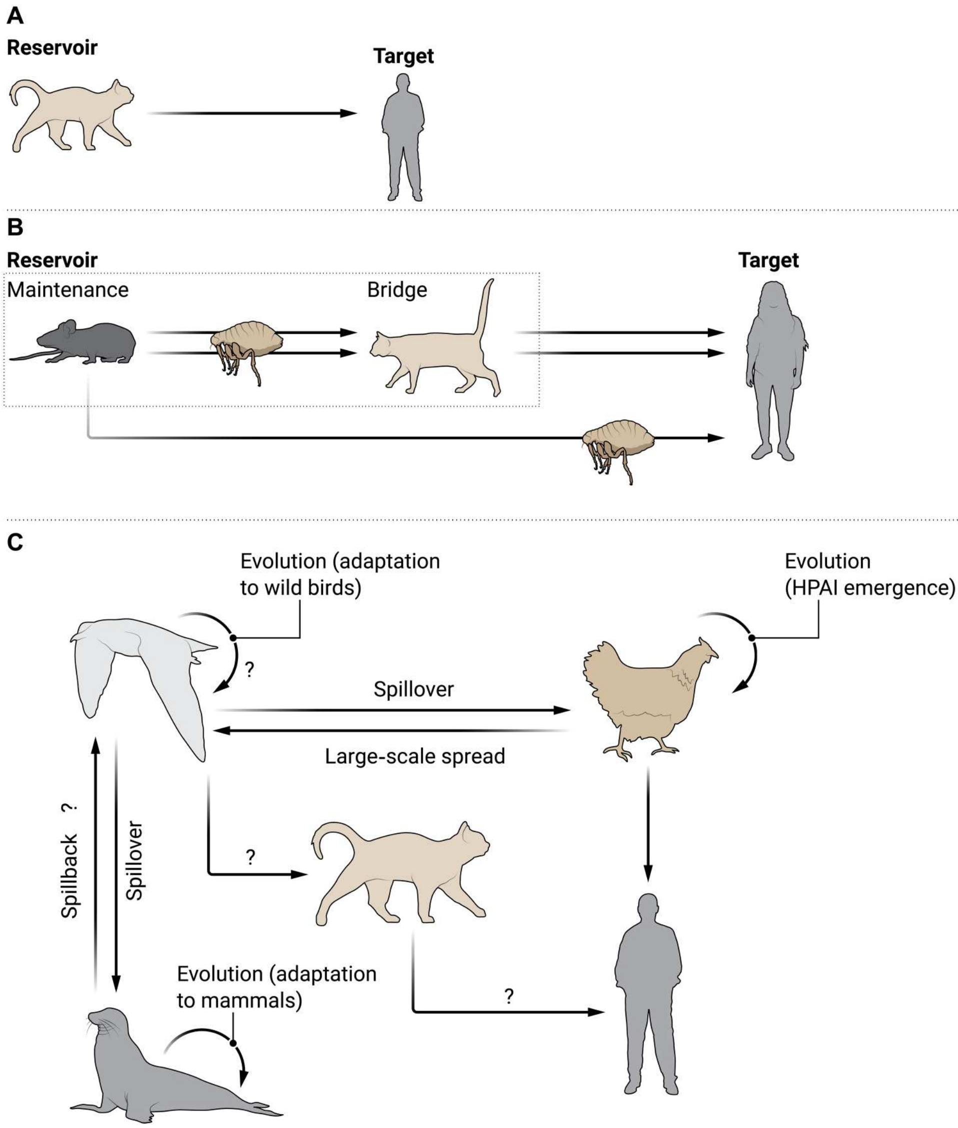

Fig. 1. Pathways for zoonotic spillovers between and via (peri-)domestic species. (**A**) Direct transmission to humans from a (peri-)domestic species acting as a reservoir (e.g., *T. gondii* in cats). (**B**) Transmission from wildlife to humans via a (peri-)domestic species acting as a bridge (e.g., *Y. pestis* in wild small mammals preyed on by cats). (**C**) More complex scenarios can include different maintenance and bridge populations, as well as populations playing specific roles in pathogen evolution (e.g., HPAI).

production, with which interactions can be particularly intimate. There are many situations by which bacteria such as *Salmonella* can act as a source of environmental infection via food sharing or pet interactions. Close contacts of feral and companion animals with wildlife also represent potential pathways for pathogen transmission from wildlife to humans, such as predation of wild rodents by cats (Fig. 2, A to C). Other situations include competition and other interactions around shared natural resources (e.g., prey competition between feral dogs and other wild carnivores) or anthropogenic resources (e.g., food scraps between backyard poultry and wild birds). There is a wide distribution of interactions, many of which have been examined through predictive modeling. Overall, there is a complex network for understanding and predicting human and animal susceptibility to infection, as illustrated by a recent review in the science of the host-pathogen network (*8*).

Molecular context

Studying the molecular factors that determine pathogen host range is critical for our understanding of pathogen transmission pathways, as well as pathogen diversity, evolution, and emergence. In the case of viruses, interactions between viruses and their host receptors mediate viral cell entry and are crucial determinants of pathogenicity, viral tropism, and host range. The determination of the tertiary structure of many virus-host receptor interfaces has revealed the molecular and biochemical interactions essential for viral entry and how mutations can affect this interaction. Thanks to the growing amount of sequence data from viruses found in different species and orthologous receptor genes from various companion and wild species, it is now possible to conduct comparative genetic analyses and explore the virus-host interaction through in silico methods to predict host susceptibility. Such studies featured

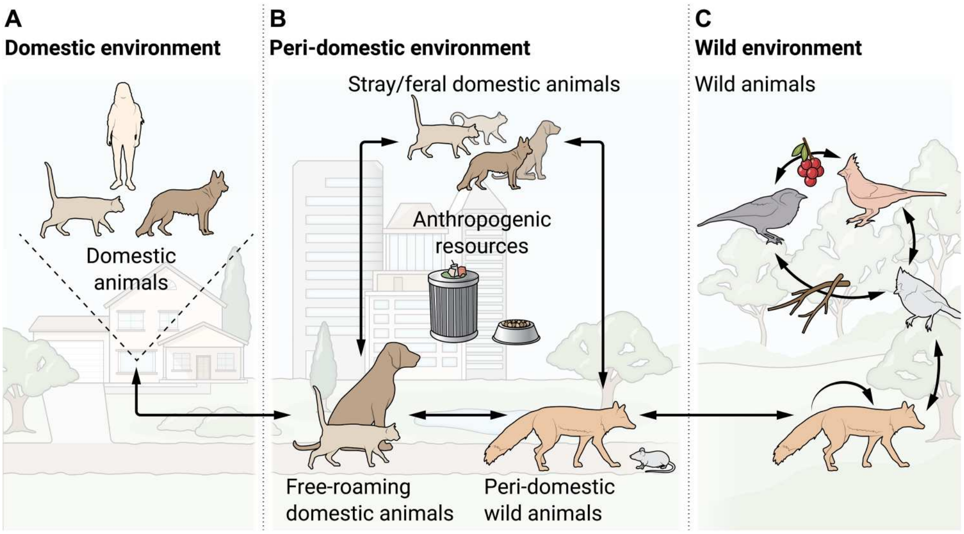

Fig. 2. Meaningful contacts between humans and domestic and wild species that can have an impact on the emergence of zoonotic pathogens. (**A**) In a household, humans have constant contact with their pets. Some households can have various species together (dogs and cats). (**B**) In urban areas, anthropogenic resources such as trash cans and feeding stations can attract various species, including human-owned free-roaming cats and dogs, stray cats and dogs, and urban wildlife (raccoons and foxes). Free-roaming pets may have various interactions with multiple species and may introduce pathogens to the house. (**C**) In rural areas, wild species are most likely to interact with individuals of their own species. Interspecific contacts are also possible through resource sharing. The extent of contact between wildlife in rural and urban areas may be limited and influenced by the increasing human encroachment in rural areas.

heavily in the early stages of the coronavirus disease 2019 (COVID-19) pandemic, although in hindsight, these may have been of limited benefit. For some multi-host viruses, such as rabies virus in bats (*9*), it has been observed that the genetic relatedness of the hosts plays an important role in virus cross-species emergence, also known as the phylogenetic distance effect (*10*), possibly because infection of related hosts requires less viral adaptation. Although the sequence of the receptor for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) [angiotensin-converting enzyme 2 (ACE2)] is highly similar across different species (*11*), host susceptibility is not necessarily determined by genetic distance, also known as the phylogenetic clade effect (*10*, *11*). Some New World primates, which have ACE2s genetically similar to humans (>92% similarity), were unable to facilitate SARS-CoV-2 cell entry (*12*). By contrast, species such as domestic cats and pigs, which are more distantly related to humans, with an ACE2 sequence similarity of less than 82%, were permissive to viral entry. Viral variants can also interact differently with receptors from different species. As another widely studied example, different variants of HIV have varying affinities to the CD4 receptor of humans (*13*) and closely related species. Macrophage-tropic HIV variants can use the CD4 of many nonhuman primate species, whereas bloodstream variants are more selective (*14*). Host population genetic variation also affects susceptibility. Different alleles of CD4 in certain chimpanzee subspecies have been detected, some of which have extra N-linked glycosylation sites. These additional glycans on CD4 impede the interaction with the chimpanzee-tropic simian immunodeficiency virus (SIVcpz) (*15*), a virus with high genetic similarity to HIV-1. Infection with SIVcpz has not been detected in subspecies of chimpanzees in which the highest prevalence of these alleles was reported (*15*) despite evidence of high exposure to virus (*16*). Spix's owl monkey (*Aotus vociferans*) populations also have different alleles of CD4, some of which can facilitate the entry of early variants of HIV-1 (*17*).

The virus-receptor interface is often a focus of spillover prediction, but other cellular factors can be of great importance. Host cell proteases are key components in determining host cell tropism and zoonotic transfer. Alterations in protease cleavage sites can readily affect host cell tropism within a host, although species-specific effects are generally not well understood. One notable exception has been described in certain zoonotic coronaviruses by which proteases can unlock barriers to infection (*18*). A number of post-entry factors can have pronounced impacts on host susceptibility. Perhaps the best known of these also involve HIV, where a variety of restriction factors (such as TRIM molecules) have been under investigation for many years (*19*). In the context of the ongoing COVID-19 pandemic, it is interesting to note the involvement of host cell factors such as IFITMs and Ly6E (*20*, *21*), which may have profound impacts on host susceptibility, acting through altered viral entry pathways and innate immune responses. It is likely that future research will identify many other restriction factors for zoonotic transmission, such as the recently found molecule BTN3A3 that plays a key role in the host range of influenza viruses (*22*).

PATHOGEN-SPECIFIC CASE STUDIES

The role of rodents as zoonotic disease reservoirs has been investigated in great detail for many years. The role of companion animals as the source of zoonotic diseases is also well acknowledged but is less well understood. For example, cat scratch fever, caused by *Bartonella henselae* infection, is a zoonotic form of bartnellosis in humans transmitted by cats (*23*). Other zoonotic pathogens commonly referred to in the veterinary literature include *C. psittaci* from pet parrots (*7*) and canine coronavirus (CCoV) in domestic dogs. However, most of those pathogens cause only rare diseases

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in humans, with relatively small impact at the population level. With companion animal numbers increasing and land-use changes creating more proximity between wildlife, people, and their companion animals, there is a need to reassess the zoonotic risks posed by (peri)domestic animals.

Dogs and cats as reservoir of rabies virus

Rabies has long been one of the most feared infectious diseases that humans contract from animals. Rabies is caused by a rhabdovirus and is invariably lethal to humans once infection is established. Dogs have historically been responsible for most rabies-associated infections from domesticated species (*24*). There has been a notable drop in dog-associated cases globally; however, despite control efforts, rabies continues to circulate at very low prevalence in several places, notably in Africa, Southeast Asia, and certain countries in South America (*25*). In the Serengeti district of Tanzania, Mancy *et al.* (*26*) have been tracking cases in dogs, as well as several other species, for more than a decade to understand the dynamics of the virus transmission. Genetic data have revealed transmission networks from which individual dog behavior emerges as a key factor. Some infected dogs may travel long distances and introduce new lineages into neighboring communities, whereas others may simply bite other animals. Dogs are social and highly mobile and have various interactions with other dogs and humans (*27*). As a result, culling has limited efficacy, with the only effective containment strategy being comprehensive dog vaccination.

Humans are not typically vaccinated for rabies, and maintenance of vaccine coverage in pets is the primary public health response, but this typically requires yearly boosters and relies on responsible ownership. Without widespread pet vaccination, feral dogs and cats, along with indoor-outdoor and free-ranging cats, are at great risk for rabies, in part because of their close proximity to wildlife hosts (e.g., raccoons, skunks, and foxes in the Americas) and relative likelihood of interactions with humans. The increase of feral cat populations and the inability of owners to afford complete veterinary care mean that rabies is one of the most concerning current zoonotic disease risks from cats, especially among free-roaming cats (*28*). Outside of the African continent, cats may now be considered a predominant rabies risk; beginning in 1988, cats were the number one domesticated species passing rabies infections to humans in the United States (*29*).

Spillovers of coronaviruses to human populations

SARS-CoV-2

At the start of the COVID-19 pandemic, after a report of SARS-CoV-2 detected in dogs (*30*), it was apparent that cats were becoming infected, based on infection of tigers and lions at the Bronx Zoo in New York City (*31*). Because of this finding and the attention garnered by testing animals at the start of the pandemic, cats were considered a high-profile risk as a reservoir of SARS-CoV-2. As the pandemic proceeded, most cases in cats have been found to be subclinical, and transmission back to humans has been confirmed (*32*) but is rare. The virus may circulate quite widely between cats, however (*33*), but the public health risk from cats is now considered low. Vaccines were developed for use in cats but never adopted for widespread use, with the exception of selected zoo animals/endangered species. As with all studies of SARS-CoV-2, it is important to assess the individual virus lineages (*34*), especially with the Omicron variant, because substantial virological changes have occurred as the pandemic has proceeded. Domestic dogs were also implicated as a public health risk, but never gained the same public profile as cats, and are also of low risk in the context of the COVID-19 pandemic.

Canine and feline coronaviruses

The COVID-19 pandemic also brought attention to the zoonotic potential of other animal coronaviruses. CCoV has recently been flagged as the eighth human coronavirus based on multiple isolations in humans (*35*, *36*). CCoV-2 belongs to the *Alphacoronavirus-1* species, together with other coronaviruses of veterinary relevance, including feline coronavirus (FCoV), transmissible gastroenteritis virus, and porcine respiratory coronavirus. Although frequently documented as a mild gastrointestinal virus in domestic dogs, highly pathogenic mutant variants of CCoV-2 have been documented, as well as several emerging recombinant variants with distinctive spike genes (*37*). CCoV-2 has also been detected in many wild species of free-ranging wild carnivores living in natural areas (*38*) and in urban settings (*39*) and in captive animals in zoos (*40*) and breeding facilities (*41*). Recently, CCoV-2 was reported in humans (*35*, *36*) and rodents (bamboo rat, *Rhizomys sinensis*) (*42*), species not shown to be infected in earlier studies. One outstanding question is whether the CCoV-2 isolated from a human with pneumonia (now known as CCoV-HuPn-2018) truly represents an example of an endemic eighth human coronavirus or just a virus isolated as a result of a spillover event. Currently, it is not known how CCoV-2 enters human cells, but a recombinant CCoV-2 recovered from a diseased dog with systemic infection was shown to be able to infect human monocyte cells in vitro (*43*). FCoV-like viruses have also been detected in humans with influenza-like symptoms in the United States (*44*). Interestingly, the S gene of one of the variants was more similar to HCoV-OC43 (a betacoronavirus) than to FCoV-1, suggesting a recombination event between these CoVs (*44*).

Other enzootic coronaviruses

It is also important to note the incidence of enzootic coronaviruses occurring in felids, which may be more substantial than appreciated. In addition to FCoV-1, there is FCoV-2 (a recombinant with CCoV-2) (*45*), as well as a deltacoronavirus in an Asian leopard cat (*Prionailurus bengalensis*) (*46*) and a newly identified luchacovirus (alphacoronavirus) in bobcats (*Lynx rufus*) (*47*). For the latter two examples, infection may be more due to cats preying on reservoir species (rodents and birds) rather than sustained cat-cat transmission. For FCoV-1, there is no well-documented study of transmission to humans, with the exception of the one study from Arkansas described above (*44*), although enzootic coronaviruses do seem to transmit easily between cats and dogs and may readily recombine. One concerning development is the increased risk of disease outbreaks due to FCoV in outdoor group-housed cats (*48*). An additional notable aspect of FCoV infection in humans is the widespread off-label use of antiviral drugs developed for SARS-CoV-2 (*49*), which is likely to involve further study and regulatory approval.

Horses are infected with an equine coronavirus (ECoV), which is a betacoronavirus related to but distinct from the bovine coronavirus (BCoV) now widely considered to be the source of the now-endemic human coronavirus (HCoV-OC43). Molecular clock analysis of HCoV-OC43 dated the common ancestor and presumed spillover event to a period around the turn of the 20th century (*50*). ECoV is an emerging infection of horses and can cause respiratory disease. Given the precedent of BCoV becoming an endemic human coronavirus and the status of horses as companion animals, ECoV should be considered a plausible pandemic-potential agent. In this context, it is noteworthy that camels can also be considered as companion animals ("beasts of burden") and are a well-recognized source of zoonotic infection based on the ongoing MERS-CoV (Middle East respiratory syndrome coronavirus) endemic outbreak in the Middle East and a viral reservoir in Africa (*51*). For the Americas, an equivalent situation may have existed for alpacas in the context of emergent alphacoronaviruses, such as HCoV-229E (*52*).

Horses as a bridge for henipaviruses

One of the most noteworthy zoonotic disease risks from horses is the situation in Australia with Hendra virus (a paramyxovirus related to Nipah virus), where horses are in contact with the bat viral vectors for the virus (flying foxes, or fruit bats of the *Pteropus* suborder). This is a situation notably linked to climate change and land use changes (*53*). The virus infects and can cause severe disease in attending veterinarians treating horses with neurological signs due to Hendra infection. Whereas case reports of zoonotic infections directly from horses remain low, there is a high potential for underreporting because of lack of knowledge among health professionals. This virus has the potential to cause greater morbidity and mortality in both rural and urban populations in the future and is of notable importance to both veterinary and human health (*54*). A vaccine is available for the equine population but has not been widely adopted (*55*).

Beyond bats, other species remain understudied as in the context of emerging paramyxoviruses, including cats who are highly susceptible to Nipah virus (*56*). Although the public health risk is currently considered to be low, this remains an area for continued surveillance.

The complex epidemiological networks of influenza viruses

Avian influenza viruses (AIVs) are typically relatively harmless to humans, and most avian infections, while readily transmitted via an enteric route, are considered of low pathogenicity (LPAI). Certain subtypes of avian influenza (H5 and H7) can become highly pathogenic (HPAI), in large part through the acquisition of a polybasic cleavage site that allows intracellular proteases, such as furin, to activate the virus. The ecological dynamics of HPAI infection have changed dramatically in recent years. Isolated epizootics of "fowl plague virus" have always been connected with poultry production, and seasonal phenomenon in migratory waterfowl has garnered wider interest based on changes in seasonality and the dramatic effects on certain wild bird populations. The increased prevalence of AIV in wild and domestic birds has also increased the risk of spillovers to humans and other mammals (*57*). AIV has also spilled over to mammalian species, mainly to wild carnivores [e.g., red foxes (*Vulpes vulpes*)] and to a variety of marine mammals in both Europe and along the Atlantic and Pacific coasts of the Americas. Infection of mink (*Neogale vision*) in fur farms is of great concern, because these high-density settings dramatically increase the likelihood of viral adaptations arising, which enable infection of mammalian hosts (*58*). Mammalian adaptation of AIVs has also been observed in wild colonial species, such as seals, which also gather in high densities (*59*). A few human AIV cases have also been reported recently, all associated to date with interactions with poultry, in particular, backyard poultry.

Mammalian companion animals represent a notable risk for influenza virus transmission to humans. Cats can be infected with human-adapted influenza viruses, likely with low public health impact. In contrast, avian-origin influenza viruses, including an H7N2 virus, may have a greater impact on transmission from cats to humans (*60*). As documented both in the past (*61*) and in more recent field reports, cats have been infected with HPAI H5N1, likely through encounters with infected birds or mammals as prey in outdoor settings. Cats appear to be highly susceptible to H5N1 (*62*), including via the gastrointestinal route, and may present with serious signs, including neurological disease. At the time of writing, concern is growing related to the detection of cats infected with H5N1 in several unconnected locations in Poland and in animal shelters in South Korea. To date, the source is unclear, but both indoor and outdoor cats are affected, raising the possibility of spread beyond the expected interaction, possibly via the human food chain or raw pet food (*63*, *64*). Influenza viruses also circulate widely in horses as a distinct H3N8 virus. The potential impact on both humans and animals was demonstrated upon the introduction of this enzootic influenza virus into the horse population of Australia, which did not have any circulating influenza viruses at that time (*65*). This situation revealed that international trade, although typically associated with production animals, can affect companion animals in certain situations. In addition, the spread of a distinct H3N2 canine influenza and its introduction into the United States via animal rescue organizations importing animals from Korea has highlighted other pathways of influenza dissemination (*66*). Improved surveillance of influenza viruses is needed and should move beyond testing of dead or sick wild animals to include a wide range of asymptomatic animals, including carnivores, because information on asymptomatic infections is currently limited (*67*).

The shifting range of vector-borne pathogens

Birds constitute the main reservoir of several zoonotic mosquito-borne viruses, including western and eastern equine encephalitis viruses and West Nile virus (WNV). These viruses can be transmitted, primarily via mosquito bites, to humans and horses, in which it can cause fatal encephalitis. With climate change and human movements causing shifts in mosquito distribution, the geographical distribution of those viruses is expected to evolve. Historically found in Eastern Africa, the range of WNV is expanding, notably because it has been spreading northward in Europe over the past decades (*68*) and through the Americas since its introduction in 1999 (*69*). The case of WNV is perhaps one of the highest-profile examples of the role of One Health in understanding a human disease outbreak. Introduction of the virus was likely human-mediated, possibly through the introduction of an infected bird or mosquito, and led to an outbreak with devastating consequences for wild birds, which acted as sentinels to detect the virus, and humans (*70*). The spread of the virus was facilitated by bird movements (*69*), which were likely facilitated by changing land use practices and climatic conditions (*71*).

Birds are also suspected of playing a role in the maintenance and spread of Lyme disease, caused by *Borrelia* spp. These are intracellular bacteria transmitted via tick bites and found in many wild and domestic species, causing both acute and chronic disease in humans. Lyme disease is one of the most notable examples of how environmental change has affected zoonotic disease and has been described as the "first epidemic of climate change." As ecological dynamics change, birds have become more prominent as alternative reservoirs. The role of birds in *Borrelia* epidemiology is still poorly understood, but the particularly high competence and tick infestation of some bird species make them a prominent alternative reservoir with wild rodents (*72*). As for other pathogens, wild birds also play a particular role in *Borrelia* epidemiology through their contribution to the large-scale spread of both ticks and *Borrelia* bacteria (*73*).

Cats as a bridge for *Yersinia pestis*

Plague is another of the most highly feared infectious diseases that humans contract from animals. Caused by the bacterium *Y. pestis*, the most common form is bubonic plague, but its pneumonic form is associated with greater mortality. The plague is historically associated with peri-domestic black rats (*Rattus rattus*), but attention has switched to other wild mammals, including prairie dogs (genus *Cynomys*) being linked to human-mediated spread of *Y. pestis* spread outside of the Old World (*74*). There is also a notable risk of transmission from cats (*75*). The agent is normally vector-borne via fleas on rodent hosts, but the pneumonic form can be aerosol-transmitted. Although aerosol transmission has been proposed as an outcome for infection of cats, the presentation in cats is primarily the bubonic form. Numbers of infections are low, but plague is endemic in 17 western U.S. states, and many of the small mammals on which cats prey carry *Y. pestis*. Consequently, outdoor cats and cats with incomplete veterinary care, combined with human interaction, suggest that cat-transmitted plague can be considered an increasing public health risk (Fig. 3). Plague is a good example of where ill-conceived responses, without input on the ecological aspects of often-complex vector-reservoir relationships, can do more harm than good, e.g., historically, the attempted removal of dead rats that precedes a human outbreak (*76*). Whereas rodents remain the notable plague reservoir, changes in the urban-rural interface and the growing role of companion animals may accelerate the incidence of this feared disease with the human populations.

Cats and the global spread of *Toxoplasma gondii*

Cats are the only definitive host for the protozoal parasite *T. gondii* (*77*), and this agent is one of the best-known examples for the risk of cats to human public health. The risk to humans is most notable during pregnancy and is often linked to litter boxes in homes, but cats only shed parasite oocysts for a week or two, and transmission risk is low with good hygiene practices. The risk to a pregnant person is more likely as a food-borne infection because *Toxoplasma* oocysts are environmentally stable and can enter the food chain. Although the contribution of cat-to-human transmission in *T. gondii* epidemiology is likely small, cats play a key role in the maintenance and spread of the parasite.

Outside of domestic situations, human-associated cat movements in particular are likely responsible for the global spread of Old World strains of *T. gondii* (*78*). The parasite is known to infect hundreds of warm-blooded species, with some showing severe neurological outcomes and threatening fragile populations such as Hawaiian monk seals (*Neomonachus schauinslandi*) (*78*). As with other notable infectious diseases, limiting feral cat populations is key to managing public health and environmental risk.

Enterobacteria and emerging antimicrobial resistance

Salmonella and *Campylobacter* bacteria are the leading causes of zoonotic diseases in humans. Those enterobacteria can cause gastrointestinal disease, especially in small children. Birds constitute an important reservoir of several *Salmonella* spp. and *Campylobacter* spp. Transmission to humans occurs primarily via the consumption of contaminated food or water and thus primarily involves domestic poultry products. Transmission from wild birds to humans also likely happens via contacts with contaminated surfaces (*79*), although the occurrence of such events remains unclear (*80*). Wild birds might also play an important role in the maintenance and spread of those bacteria, for instance, by connecting landscapes [e.g., urban units (*81*)].

As with any other bacteria, there is growing concern regarding the potential for antimicrobial resistance (AMR) in *Salmonella* spp. and *Campylobacter* spp. Various AMR genes are found in high prevalence in enterobacteria isolates from poultry (*82*) and wild birds (*83*). AMR has been mostly studied in farmed animals subject to direct antimicrobial treatment, hence a likely source of most AMR. AMR is also found in a wide range of wild bird species, with higher prevalence in those foraging on anthropogenic resources (e.g., livestock carcasses and landfills) such as gulls and vultures (*84*). Whereas AMR most likely emerges in anthropogenic environments, wild birds might contribute to their spread across landscapes (e.g., landfill and rivers) (*85*).

TOWARD A ONE HEALTH APPROACH

Tracking changes in dynamics in a changing environment

Globalization and domestic species movements have led to increased contacts with wildlife, including more and more peri-domestic populations with increased urbanization (e.g., gulls and squirrels). There are also shifts in host and pathogen distribution with global change. In this context, it is critical to implement surveillance programs allowing us to track changes in pathogen dynamics.

Companion animals and peri-domestic species can be sentinels for many zoonotic pathogens. For instance, dogs can be used as a sentinel for the emergence of *Coccidiodes* species, causing valley fever in humans. A recent study of the spatial distribution of coccidioidomycosis in dogs within California resulted in the generation of a risk map that correlated very closely with reported human cases (*86*). In particular, surveillance can facilitate the early detection of spillover events, as was the case for the WNV outbreak in New York City in 1990 (*70*). Early detection is crucial to the assessment of risk earlier in the zoonotic spillover chain of events.

In addition to case surveillance, serological surveillance in both humans and animal populations can reveal a number of potential and realized spillover events (*87*). This points us toward potential reservoir and bridge populations, which can be used to improve genetic surveillance and provide insights into transmission chains and spread, especially through application of machine learning models (*88*). Ultimately, these findings can provide information linking genotype to phenotype and allowing an assessment of pandemic risk.

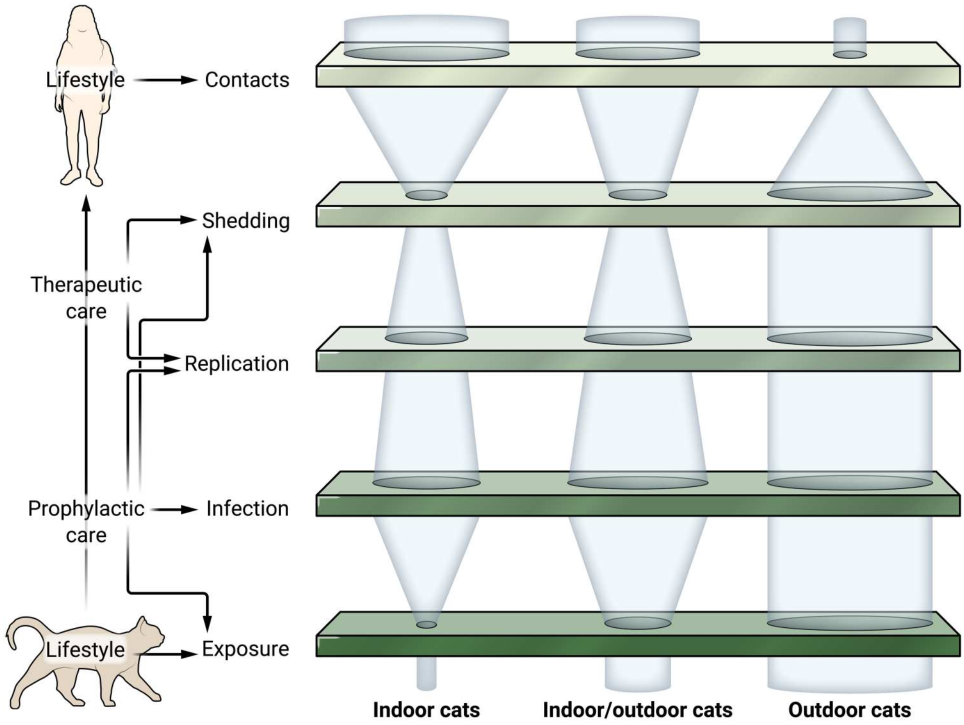

Fig. 3. Acting on transmission bottlenecks to mitigate zoonotic spillovers through (peri-) domestic species: Illustration with *Y. pestis*. Whether cats live indoors, outdoors, or in between modulates their risk of exposure and subsequent transmission of *Y. pestis*. Outdoor cats are more likely to encounter wild reservoirs (small mammals) or arthropod vectors (fleas) of the pathogen and, hence, to be exposed to the pathogen. They also usually have limited access to health care and are thus less likely to receive prophylactic care (arthropod repellents) or therapeutic care (diagnosis of the infection and subsequent antibiotic treatment), making them more likely to develop infection and shed infectious forms of the pathogens. However, indoor cats have more frequent and closer contacts with people, representing more opportunities for cat-to-people pathogen transmission.

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Acting on animal health to reduce zoonotic disease burdens in humans

There is a need to act on animal health and management to protect human health. The Guinea worm *Dracunculus medinensis* is a good example of how neglecting the epidemiological role of domestic species can hinder control. This parasitic nematode has long been a notable burden to global public health and has been targeted for eradication, with elimination achieved in many locations primarily through human case surveillance and facilitated access to clean water (*89*). However, as reports of human cases declined, domestic dogs became recognized as a reservoir (*90*) and are now being targeted for elimination of the parasite from the complete host species community.

The above case studies point toward several solutions depending on the targeted pathogen and the social and ecological contexts in which it occurs. Cats present a notable risk for zoonotic infections and present unique challenges. In addition to the examples described above, other infectious agents that can cause disease in humans include certain parasitic roundworms (*Toxocara cati*) and fungal pathogens such as *Sporothrix brasialiensis*. The zoonotic risk increases dramatically when cats are outdoors, especially when veterinary care is limited or nonexistent (e.g., feral cat populations). The role of free-roaming cats in disease ecology is perhaps the most controversial of any species, as recognized by the American Association of Feline Practitioners (AAFP). The management of free-roaming cats is complex (*91*) and includes owned cats with outdoor access, owned cats that have become lost outdoors (strays), and unowned (feral) neighborhood cats that are often supported by caregivers but are not claimed as personal pets or property. The AAFP encourages collaboration between humane groups, conservationists, animal control authorities, caregivers, and other stakeholders. In addition to domestic cats, peri-domestic felines now also include defined populations of nondomestic cats that have responded to urbanization [e.g., cougars (*Puma concolor*) that have established themselves in the Santa Monica mountains of metropolitan Los Angeles, CA] and represent a distinct risk for zoonotic infection (*92*). More interdisciplinary research is clearly needed to increase evidence-based management practices on these topics.

Good health care for domestic animals is a key part of the solution, including maintenance of vaccination protocols. We note the recent case study and widespread domestic dog rabies vaccination campaigns in Tanzania (*93*) that show that these dogs are not "feral." This scenario is also applicable to the more-established domestic situations. There is a need to maintain vaccination of pets in the face of increasing costs of veterinary care and with the ongoing threat of human/pediatric vaccine hesitancy and denial, which is becoming more prevalent. Misinformation spills over into animal scenarios, which can dramatically increase risk. Many parasitic and tick- or flea-borne infections can also be prevented with prophylactic antiparasitic treatment. Those methods can reduce both pathogen prevalence locally and pathogen spread via animal movements.

Whereas the food chain is always a risk, extensive surveillance and monitoring occurs for the human food chain. One area of note is the indirect but increasing risk of infection via "raw" pet food, including the possibility of *Mycobacterium bovis* infections causing clinical tuberculosis in humans (*94*). Raw pet food may contain many bacterial species AMR genes, as well as parasites.

Solutions must also include directly limiting or managing people's exposure to animals carrying zoonotic pathogens. For pathogens involving a domestic bridge species (e.g., *Y. pestis* via cats and henipaviruses via horses), exposure risk can be indirectly reduced by limiting contacts between the bridge species and the wild maintenance species. This can include keeping pets indoors or in confined areas (e.g., "catios"), appropriate enrichment and diet, or using collar-mounted devices impairing predation (*95*).

Acting on wildlife health is clearly more challenging, but solutions exist. In the case of pathogens circulating between humans, production animals, and wildlife, mitigation measures can include limiting wildlife access to pathogen sources such as production facilities and landfills (*83*). Limiting contacts between humans and wildlife is also possible by limiting at-risk behaviors, such as backyard feeding, or providing dedicated areas to peri-domestic species such as artificial breeding sites for birds so as to divert them from buildings used by people. Population control is another way to act on pathogen prevalence but is often counterproductive because of unexpected cascading effects. For instance, culling can lead to individual dispersal or replacement (*96*–*98*). Other solutions include neutering (*99*) or "test and vaccinate or remove" (*100*) campaigns.

Perspectives: Looking at neglected species

The response for limiting zoonotic transmission by companion animals is very different from those by livestock (typically depopulation) and wildlife. Management for companion animals includes targeted vaccination and anti-infectives (with awareness of AMR issues), and management for peri-domestic animals includes clear communication. Climate change will profoundly affect human health, leading to a new discipline of "climate epidemiology" (*101*). Whereas changes in climate conditions are most likely expected to affect the distribution of disease vectors, such as mosquitos and ticks, changes in water temperature, precipitation, and flooding may influence water-related diseases and the reservoirs for more direct zoonoses, as already demonstrated for *Leptospira*, via populations of reservoir hosts (*2*). Increased urbanization and changes in behavioral aspects of companion animal ownership will also have profound effects on disease management. These factors will fundamentally affect the risk of emerging zoonoses.

The wildlife reservoir of zoonotic diseases in bat species (order Chiroptera) is well known and prodigious and includes high-profile pathogens such as Ebola virus, Nipah virus, and rabies virus, along with the potential for emergence of previously unknown human coronaviruses. From a peri-domestic perspective, bats only rarely come into direct contact with humans, but other terrestrial mammals are more notable in this regard, especially in the context of increased urbanization and climate change. In the order Rodentia, one of the most well-characterized zoonosis examples are hantaviruses, where climate fluctuations affect rodent reservoirs. One of the most infamous of the hantaviruses is Sin Nombre virus, which emerged in the Four Corners region of the United States in 1993. These viral zoonoses occur globally and on a regular basis and include arenaviruses that are the cause of "Old World" and "New World" hemorrhagic fevers (*102*). Neglected species in the context of pathogen monitoring and surveillance that should also be considered are part of the order Eulipotyphla. This order includes shrews and hedgehogs, and although they are not a well-recognized source of zoonotic diseases, they have been shown to be reservoirs of coronaviruses that are comparable to those in bats (*103*). These animals may be kept as exotic pets or have other peri-domestic wildlife interactions with humans, thus increasing the risk of zoonotic transmission. In the future, additional neglected species may also emerge as zoonotic risks.

In summary, it is becoming well recognized that certain infectious diseases will become more problematic as a result of climate change (*104*, *105*). The impacts from companion animals are no exception to this, and wildlife reservoirs may be especially impactful in the context of new habitats based on increased urbanization and changes in human behavior. Better surveillance and monitoring of key species (e.g., cats, dogs, and horses) as well as well-reasoned implementation of control measures will be critical to our response to new risks.

CONCLUDING REMARKS

Because of their close proximity with people, companion animals and peri-domestic wildlife occupy a key position in the epidemiological networks of many zoonotic pathogens. Surveillance of those zoonotic pathogens should include companion animals and peri-domestic wildlife, and research should combine ecological approaches with molecular approaches to understand their roles as epidemiological reservoirs (e.g., dogs for rabies viruses) or bridges (e.g., horses for bat-borne paramyxoviruses). It is critically important to distinguish situations where pathogens frequently spill over from their reservoir to humans (e.g., rabies virus) versus situations where spillovers are followed by sustained transmission in humans (e.g., SARS-CoV-2). Both situations can be associated with significant public health burden but call for different control measures (*106*). Designing efficient control measures also requires accounting for the diversity of animal populations involved in those epidemiological networks because feral populations should not be overlooked over pet populations. Last, it is important to acknowledge that epidemiological networks, as well as other ecological networks, are dynamic and are expected to rapidly change with anthropogenic global change.

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Acknowledgments: We thank N. Andre for helpful comments. **Funding:** Work in the Whittaker laboratory is funded in part by the Cornell Feline Health Center. **Author contributions:** G.R.W. conceptualized the text, and A.G. and X.A.O.-C. conceptualized the figures. G.R.W., A.G., and X.A.O.-C. wrote the first draft and revised the text and figures based on reviewer and editorial comments. **Competing interests:** The authors declare that they have no competing interests.

Submitted 17 July 2023
Accepted 28 September 2023
Published 18 October 2023
10.1126/scitranslmed.adj0037

Backyard zoonoses: The roles of companion animals and peri-domestic wildlife

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Sci. Transl. Med. **15** (718), eadj0037. DOI: 10.1126/scitranslmed.adj0037

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